Day 0

Installation and Setup

SQLCourse: 30 days from beginner to job-ready SQL

Goal: Complete this day before Day 1. Estimated time: 45-90 minutes. By the end you will have a working database, a tool to run queries, and a successful first query.

Learning objectives

- Install a SQL database system and a tool to write queries on your computer.
- Create the **sqlcourse** database that you will use for all 30 days.
- Run your first query and confirm everything works.
- Know how to fix the most common setup problems.

Choose your path

This course uses **PostgreSQL**, a free, professional-grade database used by many companies. You will control it with **pgAdmin**, a free graphical tool. If your computer is old, restricted (for example a locked work laptop), or the PostgreSQL install keeps failing, use the lighter **SQLite** path instead. Everything in the course runs on both.

	Path A: PostgreSQL + pgAdmin (recommended)	Path B: SQLite + DB Browser
Best for	Learners who want job-realistic skills	Older or restricted computers, or a quick start
Install effort	Medium (installer with a few screens)	Very low (one small app)
Needs a server running	Yes (starts automatically after install)	No (a database is just a file)
Course compatibility	Full	Full (a few Phase 4 topics differ; we will flag them)

Pick one path now. You can always add the other later. Do not install both at the same time while troubleshooting.

Path A: PostgreSQL + pgAdmin

Before you start: have your computer's admin password ready, and about 1 GB of free disk space. **Write down the password you choose for the database in Step 3 and keep it safe.** You will need it every time you connect.

A1. Windows

- Go to **postgresql.org/download**, choose **Windows**, then follow the link to the installer from EDB. Download the latest stable installer for your system (64-bit for nearly all computers).
- Run the downloaded file. If Windows asks for permission, click Yes.
- Click Next through the first screens. On the **Select Components** screen, keep these ticked: **PostgreSQL Server**, **pgAdmin 4**, and **Command Line Tools**. (Stack Builder is optional; you can untick it.)
- Keep the default install and data folders.
- **Password screen:** choose a password for the database superuser, named **postgres**. Pick something simple you will remember, and write it down.
- **Port screen:** keep the default **5432**. Locale: keep the default.
- Click Next, then Install. It takes a few minutes. Finish the wizard.
- Open the Start menu and search for **pgAdmin 4** to confirm it is installed.

A2. macOS

Easiest method: the same graphical installer from EDB.

- Go to **postgresql.org/download**, choose **macOS**, follow the link to the EDB installer, and download it.
- Open the downloaded file and follow the same screens as Windows: keep PostgreSQL Server, pgAdmin 4, and Command Line Tools ticked, set a password for the **postgres** user, keep port **5432**.
- When finished, open **pgAdmin 4** from your Applications folder.

Alternative for Homebrew users: run the following in Terminal, then install pgAdmin from **pgadmin.org/download**. Check the command output for the exact service name your install uses.

```
brew install postgresql
brew services start postgresql
```

A3. Linux (Ubuntu / Debian)

- Open a terminal and install PostgreSQL:

```
sudo apt update
sudo apt install postgresql postgresql-contrib
```

Check that the service is running:

```
sudo systemctl status postgresql
```

Set a password for the **postgres** user so you can log in from pgAdmin:

```
sudo -u postgres psql
ALTER USER postgres PASSWORD 'choose_a_password';
```

```
\q
```

Install pgAdmin by following the Linux (APT) instructions at **pgadmin.org/download**. They change over time, so use the page rather than copying commands from memory. Other distributions (Fedora, Arch, and so on) use their own package manager: search for "install postgresql on <your distro>".

A4. Connect pgAdmin to your server

- Open **pgAdmin 4**. The first time, it asks you to set a **master password**. This protects saved passwords inside pgAdmin and is separate from your database password.
- In the left panel, expand **Servers**. If you see **PostgreSQL** there, double-click it and enter the database password from the install. Tick *Save Password* if offered.
- If no server is listed: right-click **Servers**, then **Register**, then **Server**. On the General tab, Name: **Local PostgreSQL**. On the Connection tab: Host **localhost**, Port **5432**, Maintenance database **postgres**, Username **postgres**, and your password. Click Save.
- A green/active server icon with databases listed beneath it means you are connected.

A5. Create the sqlcourse database

- In pgAdmin, right-click **Databases**, then **Create**, then **Database**.
- Database name: **sqlcourse** (all lowercase). Leave Owner as **postgres**. Click Save.
- Click **sqlcourse** in the left panel to select it, then open **Tools**, then **Query Tool**. A blank editor opens. This is where you will write all your SQL.
- Check the title of the Query Tool tab or the connection shown at the top. It must mention **sqlcourse**. If it says postgres, you are in the wrong database.

Prefer text? Equivalent in SQL (run it while connected to the default **postgres** database):

```
CREATE DATABASE sqlcourse;
```

Path B: SQLite + DB Browser for SQLite

SQLite needs no server and no password. A database is one file on your computer.

B1. Install

- Go to **sqlitebrowser.org**, open the Downloads page, and pick the installer for your system (Windows, macOS, or Linux packages are provided).
- Install it like any normal app and open **DB Browser for SQLite**.

B2. Create the sqlcourse database

- Click **New Database**.
- Save the file as **sqlcourse.db** in a folder you will remember, for example Documents/SQLCourse.
- A dialog may open asking you to create a table. Click **Cancel**; we will create tables on Day 1.
- Open the **Execute SQL** tab. This is your query editor. Press **Ctrl+Enter** (Cmd+Return on Mac) or click the play button to run.

Important: Each time you return, use **Open Database** and choose your sqlcourse.db file. Remember to click **Write Changes** (Ctrl+S) to save when you create or change data. Day 1 will remind you.

Optional: the command-line tool

Many systems already include the **sqlite3** command (macOS does; most Linux systems do). Test it in a terminal:

```
sqlite3 sqlcourse.db
```

Type **.quit** to leave. You do not need this for the course.

Your first queries (both paths)

Type each query into your query editor exactly as shown, then run it. In pgAdmin, press **F5** or click the play button. A SQL statement ends with a semicolon.

Query 1: say hello

```
SELECT 'Hello, SQL!' AS greeting;
```

You should see one row with one column named **greeting**. If so, your editor works.

Query 2: do some maths

```
SELECT 2 + 3 AS result, 10 * 4 AS product;
```

Expected: result = 5, product = 40.

Query 3: check your version

PostgreSQL path:

```
SELECT version();
```

SQLite path:

```
SELECT sqlite_version();
```

Both return one line of text telling you which version you are running. Note it down; if you ever need help, include it.

Query 4: today's date

PostgreSQL path:

```
SELECT CURRENT_DATE;
```

SQLite path:

```
SELECT DATE('now');
```

> **Note:** The two systems differ slightly in places like this. Whenever it matters, the course will show both versions. Everything that is the same will be shown once.

Reading the result

The result appears as a grid below the editor. Each **column** has a name and each **row** is one record. You will learn all about this on Day 1.

Practice checkpoints

Day 0 has no puzzles, only a short checklist to prove your setup is ready. Complete each one.

Checkpoint 1 (easy)

Run a query that returns your own name as a column called **my_name**.

Checkpoint 2 (easy)

Run a query that calculates the number of minutes in a week and shows it as **minutes_in_week**.

Checkpoint 3 (easy)

Run a query that returns two columns in one row: your favorite number and your favorite word.

Checkpoint 4 (medium)

Run a query that deliberately has a typo, for example **SELEC 1;**. Read the error message. Then fix the typo and run it again. Learning to read errors is a core skill.

Checkpoint 5 (medium)

Close your query tool completely (pgAdmin or DB Browser), reopen it, reconnect to **sqlcourse**, and run Query 1 again. If you can do this without help, you are ready for Day 1.

Stretch goal

Run a query with several calculations at once, for example the number of seconds in a day, the area of a circle of radius 7 using 3.14159, and your age in months. Give every column a clear name using **AS**.

Setup checklist

Item	Done?
Database system installed (PostgreSQL or SQLite)	
Query tool opens without errors (pgAdmin or DB Browser)	
Database named sqlcourse (or sqlcourse.db) exists	
Hello, SQL! query returned a result	
Database password written down (PostgreSQL path)	

Recap

One-line takeaway: a database system stores your data, a query tool lets you talk to it, and SQL is the language you speak. You now have all three.

Common mistakes

- Forgetting the database password. Without it you must reset it, so write it down at install time.
- Running queries while connected to the wrong database (for example postgres instead of sqlcourse).
- Typing curly or "smart" quotes copied from a document. SQL needs straight quotes: single quotes **'like this'** for text.

- Forgetting the semicolon at the end of a statement.
- Using SQLite and forgetting **Write Changes** after changing data.

Troubleshooting

Problem	What to try
Installer fails or hangs (Windows)	Right-click the installer and choose Run as administrator. Temporarily pause antivirus. Make sure you have enough disk space. Try again.
pgAdmin shows "password authentication failed"	You are using the wrong password. Use the password you set for the postgres user during install, not the pgAdmin master password. On Linux, reset it with the ALTER USER command from step A3.
"Could not connect to server" or "connection refused"	The PostgreSQL service is not running. Windows: open Services, find postgresql, and Start it. Linux: sudo systemctl start postgresql. Also check the port is 5432.
Port 5432 already in use	Another PostgreSQL is installed. Uninstall the old one, or choose a different port (for example 5433) during install and use it in pgAdmin.
pgAdmin opens but is very slow to start	This is normal on the first launch. Wait a minute. Browser-based pgAdmin may open in your web browser; keep the window open.
I cannot find the Query Tool	Select the sqlcourse database in the left panel first, then open Tools and choose Query Tool.
"database sqlcourse does not exist"	Create it using step A5. Names are case-sensitive in some setups, so use lowercase.
DB Browser cannot open my file (SQLite)	Check the file is named sqlcourse.db and is not stored in a protected folder. Use Open Database, not New Database.
Mac says the app cannot be opened (unidentified developer)	Open System Settings, then Privacy and Security, and choose Open Anyway for the app.
Nothing works and I am stuck	Switch to Path B (SQLite). You lose nothing for the course.

Getting help: Copy the exact error text and search for it, or paste it into a chat with the version you noted earlier. Error messages are very specific and usually point to the fix.

Answer key with explanations

Checkpoint 1

```
SELECT 'Raj' AS my_name;
```

Any name works. Text is wrapped in single quotes. **AS** gives the column a label. (If you used double quotes around the text, PostgreSQL will read it as a column name and fail.)

Checkpoint 2

```
SELECT 7 * 24 * 60 AS minutes_in_week;
```

Result: 10080. SQL calculates numbers directly; no table is needed for this.

Checkpoint 3

```
SELECT 7 AS favorite_number, 'blue' AS favorite_word;
```

Separate columns with a comma. Numbers need no quotes; text does.

Checkpoint 4

```
SELEC 1;      -- error
SELECT 1;     -- fixed
```

The error will say something like *syntax error at or near "SELEC"*. It names the exact spot where SQL got confused. Fix the spelling and rerun.

Checkpoint 5

No code. You should be able to reopen the tool, pick the sqlcourse database, and run a query without help. If reconnecting asks for a password, that is expected on the PostgreSQL path.

Stretch goal (one possible answer)

```
SELECT
  24 * 60 * 60 AS seconds_in_day,
  3.14159 * 7 * 7 AS circle_area,
  30 * 12 AS age_in_months;
```

Replace 30 with your own age. One query can compute many things at once; just separate each with a comma and name each with AS.

Next up: Day 1 introduces what a database really is, how tables, rows, columns and keys work, and loads the course dataset (customers, products, orders, order_items) into your new sqlcourse database. Bring your working setup.